

LESSON 3-5 • HOMEWORK PACKET • SHEET 1 OF 2**Comparing Ratios with Common Terms**

Maryland Standard 6.RP.A.3b • Learning target – I can compare two ratios by scaling them until they share a common term.

REMEMBER

- To compare two ratios, scale them so one term is the **same**. Then compare the other terms.
- A bigger first number alone does **not** tell you which mix is stronger.

WORKED EXAMPLE

Punch A uses 2 cups juice to 3 cups water. Punch B uses 3 cups juice to 5 cups water. Which tastes more like juice?

Step 1 Match the juice (LCM of 2 and 3 is 6): A = 6 : 9, B = 6 : 10

Step 2 Same juice, and A has less water

Answer **Punch A tastes more like juice.**

PRACTICE

show your work • label units

1. Scale each ratio so the first term is 24.

a. $3 : 5 = 24 : \underline{\hspace{2cm}}$

b. $4 : 7 = 24 : \underline{\hspace{2cm}}$

c. $6 : 11 = 24 : \underline{\hspace{2cm}}$

d. $8 : 9 = 24 : \underline{\hspace{2cm}}$

2. Mix P uses 3 cups blue paint to 4 cups yellow. Mix Q uses 5 cups blue to 6 cups yellow. Complete the tables. Which mix is bluer?

P: blue	3	6	9	12	15
P: yellow	4				
Q: blue	5	10	15	20	25
Q: yellow	6				

Bluer mix _____

3. **Garden Seeds.** In Pack A, 9 out of every 12 seeds sprout. In Pack B, 13 out of every 16 seeds sprout.

a. Scale Pack A to 48 seeds: _____

b. Scale Pack B to 48 seeds: _____

c. Which pack sprouts better? _____

4. **Find the Error.** Drink X uses 4 spoons sugar to 9 cups water. Drink Y uses 6 spoons sugar to 15 cups water. Jamal says Y is sweeter because $6 > 4$. Is he right? Explain.

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____

AT HOME Mix two glasses of a drink with different amounts. Ask someone at home which is stronger and why.

LESSON 3-5 • HOMEWORK PACKET • SHEET 2 OF 2**Comparing Rates with Benchmarks**

Maryland Standard 6.RP.A.3b • Learning target – I can use a benchmark to compare two rates without dividing.

REMEMBER

- A **benchmark** is a friendly amount both rates can reach.
- Scale both rates to the benchmark, then compare.
Same amount: **lower price** wins.

WORKED EXAMPLE

Which is the better buy: 4 cans for \$3 or 6 cans for \$4.80?

Step 1 Benchmark: 12 cans (4×3 and 6×2)**Step 2** 12 cans: $\$3 \times 3 = \9.00 vs. $\$4.80 \times 2 = \9.60 **Answer** 4 cans for \$3 is the better buy.**PRACTICE***show your work • label units*

- Name a benchmark amount both rates can reach.
 - a 3-pack and a 4-pack _____
 - 6 minutes and 9 minutes _____
- Kai types 90 words in 2 minutes. Lena types 140 words in 3 minutes. Circle the faster typist.

	Minutes	Words	Words in 6 minutes
Kai	2	90	
Lena	3	140	

- Bird Seed.** Store A: 5 pounds of bird seed for \$8. Store B: 8 pounds for \$12. Use a 40-pound benchmark.
 - Cost of 40 pounds: Store A _____ Store B _____
 - Which store is the better buy? _____
- Trail Mix.** Store C: 3 lb of trail mix for \$7. Store D: 4 lb for \$9. Use a 12-lb benchmark.
 - Cost of 12 pounds: Store C _____ Store D _____
 - Which store is the better buy? _____
- Reasoning.** Car 1 goes 150 miles on 5 gallons. Car 2 goes 200 miles on 8 gallons. Which is more efficient? Use a benchmark.

TEST PRACTICE*choose the best answer*

- Which benchmark could you use to compare 6 bagels for \$4 and 8 bagels for \$5?

A 14 bagels **B** 24 bagels **C** 16 bagels **D** 10 bagels
- Dante walks 3 mi in 50 min. Ella walks 4 mi in 60 min. Using a 12-mile benchmark, who is faster?

A Dante (200 min) **B** Ella (180 min) **C** Dante (150 min) **D** Same speed

CHECK YOURSELF

- ☐
- I can do this on my own
- ☐
- I need a little help
- ☐
- I have a question about #_____

AT HOME Compare two sizes of one item in a store ad using a benchmark.